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Section 61

Group #7

Image Processing: Brain Tumor Classification

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Section 1

Introduction

Problem Statement

Brain tumors are life-threatening if not detected early, and accurate diagnosis is essential for effective treatment. MRI imaging provides detailed insights, but analyzing these images manually is time-consuming and requires expert radiologists. Automating tumor classification can help speed up diagnosis and improve accuracy.

Dataset

We use the Kaggle Brain Tumor Classification dataset (version 3), which contains 3,264 MRI images divided into four classes: meningioma, glioma, pituitary tumor, and no tumor. The training set includes 2,870 images, and the testing set includes 394 images. The images vary in resolution, typically between 200×200 and 512×512 pixels, and present challenges such as class imbalance and differences in tumor appearance, shape, and imaging quality.

Plan

We will train three classical machine learning models, Random Forest, Decision Tree, and Logistic Regression, using flattened and texture-based image features. In addition, we will develop a deep learning model that learns features directly from images. All models will be evaluated using accuracy, precision, recall, F1 score, and confusion matrices.

Section 2

2.0 Research Topic

Brain tumors are serious medical conditions that require early and accurate detection for effective treatment. MRI imaging provides detailed views of the brain, but manually analyzing these scans is slow and depends heavily on expert radiologists. Our research focuses on automating brain tumor classification using machine learning and deep learning techniques to support faster, more accurate diagnosis and improve clinical decision-making.

2.1 Literature Review

Study 1: Multi-class Classification of Brain Tumor Types from MR Images Using EfficientNets

Objective: Develop an automated system to classify brain tumors into glioma, meningioma, and pituitary tumors using an EfficientNet-based deep learning model.

Data: A publicly available contrast-enhanced brain MRI dataset from Figshare containing labeled images of the three tumor types.

Methods: The study used a deep convolutional neural network based on EfficientNet with pretrained ImageNet weights. Custom trainable layers and a 3-class output layer allowed the model to automatically extract features from MRI images without manual engineering. Transfer learning and EfficientNet's compound scaling improved accuracy and training efficiency, while fine-tuning higher layers preserved low-level features, enabling the network to learn meaningful patterns directly from the scans.

Key Results: EfficientNet-B2 achieved 98.86% accuracy, 98.65% precision, 98.77% recall, and 98.71% F1-score, indicating strong performance for multi-class brain tumor classification.

Reference:

Zulfiqar, F., Ijaz Bajwa, U., & Mehmood, Y. (2023). Multi-class classification of brain tumor types from MR images using EfficientNets. Biomedical Signal Processing and Control, 84, 104777. <https://doi.org/10.1016/j.bspc.2023.104777>

Study 2: A Transfer Learning Approach for AI-Based Classification of Brain Tumors

Objective: To evaluate pre-trained CNN models using transfer learning for classifying MRI brain tumors as benign or malignant, and to explore their ability to distinguish tumor subtypes and glioma grades.

Data: The study used a dataset of 696 MRI images from TCIA, including 224 benign and 472 malignant cases.

Methods: Multiple pre-trained CNN architectures were fine-tuned by adding custom layers and adjusting weights after preprocessing the MRI images. Transfer learning reduced training time and improved generalization, while fine-tuning the upper layers allowed the models to adapt to MRI-specific features. The CNNs automatically learned discriminative patterns for tumor type and grade, eliminating the need for manual feature extraction. Several architectures were compared to determine the most effective model.

Key Results: AlexNet achieved the highest accuracy at 99.04%, indicating the strong effectiveness of transfer learning for brain tumor classification.

Reference:

Mehrotra, R., Ansari, M. A., Agrawal, R., & Anand, R. S. (2020). A Transfer Learning approach for AI-based classification of brain tumors. Machine Learning with Applications, 2, 100003. <https://doi.org/10.1016/j.mlwa.2020.100003>

Study 3: Hybrid Segmentation and Machine Learning Approach for Brain Tumor Classification

Objective: Develop an accurate brain tumor diagnosis system using a hybrid approach that combines segmentation, feature extraction, and machine learning classification.

Data: A publicly available MRI dataset containing images of brains with and without tumors, covering different tumor types and sizes.

Methods: MRI images were preprocessed with noise reduction and contrast enhancement, and tumor regions were segmented using an optimized Otsu thresholding method (EFO-OTSU). Texture and structural features were extracted with DWT and reduced using PCA. A Boosted Multi-Gradient SVM (BMG-SVM) was then used for classification, combining accurate segmentation, efficient feature extraction, and a robust classifier to reliably identify tumor regions and types.

Key Results: The hybrid method achieved very high diagnostic accuracy, demonstrating that combining optimized segmentation, effective feature extraction, and a boosted SVM classifier can reliably detect and classify brain tumors from MRI images.

Reference:

Kalaiselvi, S. & Thailambal, G. (2024). Brain tumor diagnosis from MR images using boosted multi-gradient support vector machine classifier. Measurement: Sensors, 32, 101071.

<https://doi.org/10.1016/j.measen.2024.101071>

Study 4: A Hybrid Deep Learning Model for Brain Tumor Classification

Objective: Develop a hybrid system that combines deep CNN feature extraction with an SVM classifier to improve multi-class brain tumor classification.

Data: A publicly available MRI dataset containing glioma, meningioma, pituitary tumors, and normal cases.

Methods: Deep features were extracted using a pre-trained GoogleNet (Inception) model and then classified with an SVM, forming a hybrid CNN + SVM framework. This approach leveraged GoogleNet's strong feature extraction capabilities and SVM's effective decision boundaries. The study also compared this hybrid model with a fully fine-tuned GoogleNet trained end-to-end, specifically for four-class tumor classification.

Key Results: The hybrid method achieved 98.1% accuracy, 98.2% precision, and 98.1% recall, outperforming the fully fine-tuned GoogleNet model.

Reference:

Nassar, S. E., Yasser, I., Amer, H. M., et al. (2023). A Hybrid Deep Learning Model for Brain Tumour Classification - Google-Net and SVM. Entropy, 24(6), 799.

<https://doi.org/10.3390/e24060799>

Study 5: MRI Brain Tumor Image Classification Using a Combined Feature and Image-Based Classifier

Objective: Evaluate multiple models for classifying MRI brain tumors as benign or malignant and introduce a hybrid model, CFIC, that combines feature-based and image-based learning.

Data: The Brain Tumor Detection 2020 dataset from Kaggle, containing labeled MRI scans.

Methods: Seven classifiers were developed, some trained on numerical features extracted from original or segmented tumor regions using DNNs, and others trained directly on MRI images using DCNNs. Tumor-region segmentation helped focus the analysis, and the final hybrid model, CFIC, fused the outputs of both pipelines using Least-Square Fusion. This approach combined feature-based and image-based learning, automatic pattern extraction, and decision-level fusion to improve overall prediction accuracy.

Key Results: The CFIC model achieved 98.97% accuracy, 98.86% sensitivity, and 97.14% specificity, demonstrating that combining feature-based and image-based analysis enhances diagnostic performance.

Reference:

Veeramuthu, A., Meenakshi, S., Mathivanan, G., Kotecha, K., Saini, J. R., Vijayakumar, V., & Subramaniaswamy, V. (2022). MRI Brain Tumor Image Classification Using a Combined Feature and Image-Based Classifier. *Frontiers in Psychology*, 13, 848784.
<https://doi.org/10.3389/fpsyg.2022.848784>

Project Approach: From the 5 studies, we will adopt key techniques to guide our project. We will use three classical machine learning models, Random Forest, Decision Tree, and Logistic Regression, on flattened and texture-based features, alongside a deep learning model that learns features directly from MRI images. Preprocessing and data augmentation will be applied to improve image quality and reduce class imbalance, and all models will be evaluated across the four tumor classes, including “no tumor,” using accuracy, precision, recall, F1 score, and confusion matrices.

Section 3

3.0 Data Preparation

The Brain Tumor Classification dataset contains MRI images belonging to four categories: meningioma, glioma, pituitary tumor, and no tumor. Before training the models, all images were passed through a simple preprocessing pipeline to standardize their format.

Each image was loaded, converted to grayscale, resized, and its pixel values were normalized to the 0-1 range. Class labels were also encoded as integers for model compatibility:

0 = meningioma, 1 = glioma, 2 = pituitary tumor, and 3 = no tumor.

For the classical machine learning models, the images were flattened into 1-D feature vectors, allowing algorithms such as **Logistic Regression**, **Decision Tree**, and **Random Forest** to operate on the pixel values directly. For the deep learning model, the images were kept in their 2-D structure with a single channel so that spatial features could be learned.

The dataset already comes with predefined training and testing folders. All models were trained using the images in the *training* directory and evaluated using the images in the *testing* directory.

Section 4

4.0 Model Development

This section describes the development of the classical machine learning models used for brain tumor classification. All models were trained on the preprocessed images from the training set and evaluated on the testing set using accuracy, precision, recall, and F1-score.

4.1 Machine Learning Model

Three classical models were implemented: Logistic Regression, Decision Tree, and Random Forest. Their performance was evaluated using standard classification metrics:

- Accuracy: overall percentage of correct predictions.
- Precision: out of the images predicted as a certain class, how many were correct.
- Recall: out of all images belonging to a class, how many were correctly detected.
- F1-score: the harmonic mean of precision and recall, providing a balanced measure of performance.

4.1.1 ML Model 1: Decision Tree

The Decision Tree reached 72.33% accuracy. In this dataset, the model works by repeatedly splitting the pixel features into decision rules to classify an image. While this allows it to capture some patterns, the high number of pixels causes the tree to overfit, making it less reliable on unseen meningioma and no-tumor images.

```

=== Decision Tree ===
              precision    recall  f1-score   support

      0         1.00      0.21      0.35        100
      1         0.76      0.97      0.85        115
      2         0.61      1.00      0.76        105
      3         0.87      0.65      0.74         74

   accuracy                   0.72        394
  macro avg              0.81      0.71      0.67        394
 weighted avg              0.80      0.72      0.68        394

```

```

Confusion Matrix:
[[ 21  20  53   6]
 [   0 111   3   1]
 [   0   0 105   0]
 [   0  16  10  48]]
Accuracy: 0.7233502538071066

```

Figure 1: The Decision Tree shows good predictions for glioma and pituitary tumors but struggles with meningioma and no-tumor images.

4.1.2 ML Model 2: Random Forest

Random Forest achieved 69.79% accuracy. Here, the model builds multiple decision trees, each trained on different subsets of pixel features. The trees vote on the final class. While this reduces overfitting compared to a single tree, the model still struggles because flattened MRI images create very high-dimensional data, making it difficult for the trees to learn stable patterns, especially for meningioma.

```

=== Random Forest ===
              precision    recall  f1-score   support

     0         1.00      0.18      0.31      100
     1         0.83      0.97      0.90      115
     2         0.54      1.00      0.70      105
     3         0.87      0.54      0.67       74

 accuracy          0.70      394
 macro avg          0.81      0.67      0.64      394
 weighted avg       0.80      0.70      0.65      394

Confusion Matrix:
[[ 18  17  59   6]
 [   0 112   3   0]
 [   0   0 105   0]
 [   0   6  28  40]]
Accuracy: 0.6979695431472082

```

Figure 2: The Random Forest model detects glioma and pituitary tumors well but misclassifies many meningioma and no-tumor images.

4.1.3 ML Model 3: Logistic Regression

Logistic Regression achieved the highest accuracy (74.36%). In this dataset, the model learns to separate the four tumor classes by applying a set of weighted pixel features and passing them through a logistic (sigmoid/softmax) function. This creates probability-based decisions about which class an image belongs to. The model performs well when the visual differences between classes are consistent, which is why glioma and pituitary tumors are identified accurately. However, meningioma images share more similarities with other classes, making them harder for the model to distinguish.

```

=== Logistic Regression ===
              precision    recall  f1-score   support

     0         1.00      0.26      0.41       100
     1         0.74      0.98      0.85       115
     2         0.66      1.00      0.80       105
     3         0.86      0.66      0.75        74

 accuracy                   0.74       394
 macro avg              0.82      0.73      0.70       394
 weighted avg           0.81      0.74      0.70       394

Confusion Matrix:
[[ 26  23  43   8]
 [   0 113   2   0]
 [   0   0 105   0]
 [   0  16   9  49]]
Accuracy: 0.7436548223350253

```

Figure 3: The Logistic Regression model correctly identifies most glioma and pituitary tumors but misclassifies many meningioma and some no-tumor images.

4.2 Deep Learning Model: MobileNetV2

A deep learning model was developed using MobileNetV2, a convolutional neural network pretrained on the ImageNet dataset. Using a pretrained model applies the concept of *transfer learning*, where knowledge learned from a large, general dataset is reused for a smaller, specific dataset (in this case, the brain tumor classification dataset). This allows the model to benefit from previously learned visual features such as edges, textures, and shapes.

In this project, MobileNetV2 was used as a *feature extractor*, and the final layers were *fine-tuned* on the brain tumor MRI training set. Unlike the classical ML models, which rely on flattened pixel values, the CNN analyzes the images in their 2-D form, making it capable of learning spatial tumor characteristics.

The model was trained with standard preprocessing (resizing, normalization, batching) and optimized with early stopping to prevent overfitting. This deep learning approach achieved more accurate and balanced results across all classes compared to the classical models.

Final Test Accuracy (MobileNetV2): 0.8350253701210022

Figure 4: MobileNetV2 achieved a higher test accuracy of 83.50%, demonstrating stronger and more consistent performance across all classes.

Section 5

5.0 Model Evaluation

Comparison Between the Machine Learning Models

Among the three classical models, Logistic Regression performed the best, followed by Decision Tree, then Random Forest. Logistic Regression achieved higher accuracy because it is more stable on high-dimensional data; even though images were flattened, the model could still learn useful pixel-intensity patterns. In contrast, Decision Trees and Random Forests struggled because tree-based models do not handle large numbers of continuous features well. With thousands of pixels per image, the trees tend to create overly complex rules, leading to overfitting and poor generalization. This explains why both tree-based models misclassified many meningioma and no-tumor images, which have more subtle differences.

ML vs. DL Performance

The deep learning model (MobileNetV2) outperformed all classical models, achieving $\approx 84\%$ accuracy compared to ML's best performance model (Logistic Regression with $\approx 74\%$ accuracy). This improvement is expected because CNNs learn spatial features directly from the image, such as tumor shape, texture, and edges, patterns that flattened ML inputs cannot capture. Additionally, using *transfer learning* from ImageNet allows MobileNetV2 to start with pretrained feature detectors, making it much better at identifying complex visual structures in MRI scans. As a result, the DL model shows more balanced performance across all four classes and reduces the misclassification seen in the ML models.